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Network and System Admin

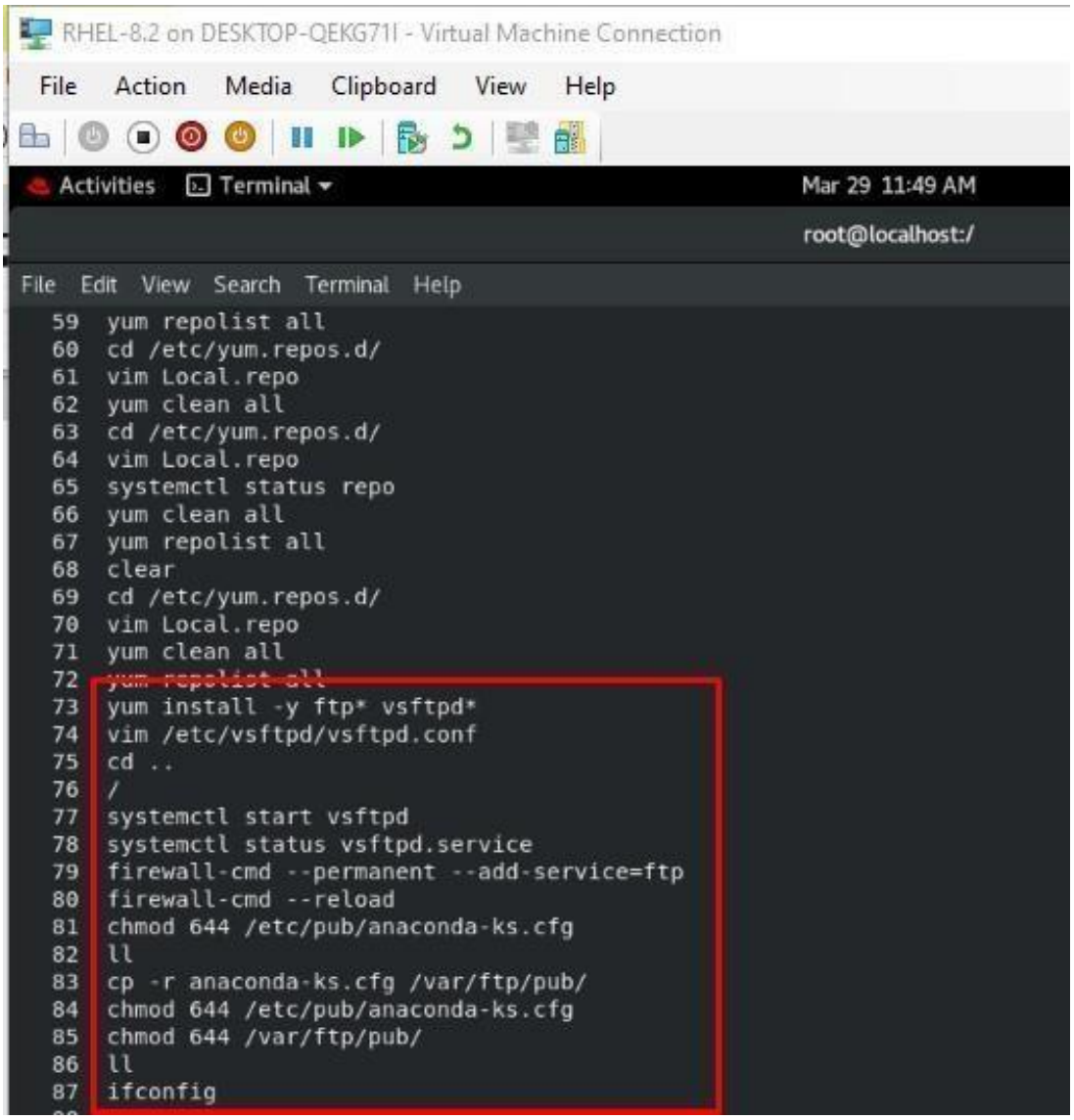
RED HAT ENTERPRISE LINUX

Kickstart, Cockpit and KVM , Samba Implementation

1. Automated Installation via Kickstart

Objective

The goal was to automate the RHEL 8 installation process to ensure consistency and save time when deploying multiple servers.



Implementation Steps -

- **Service Configuration**: Installed and enabled the **vsftpd** package to host the configuration files.
- **File Deployment**: Copied the default installation script (anaconda-ks.cfg) to the /var/ftp/pub/ directory.
- **Permissions**: Applied chmod 644 to the Kickstart file to ensure it is readable by the installer over the network.

.

```
[root@localhost /]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.3.62  netmask 255.255.255.0  broadcast 192.168.3.255
        inet6 fe80::6809:a5b:8d49:eb4a  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:03:3d:00  txqueuelen 1000  (Ethernet)
        RX packets 18022  bytes 3976652 (3.7 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 678  bytes 44450 (43.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 5152  bytes 8128258 (7.7 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 5152  bytes 8128258 (7.7 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.122.1  netmask 255.255.255.0  broadcast 192.168.122.255
        ether 52:54:00:c7:8f:1b  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

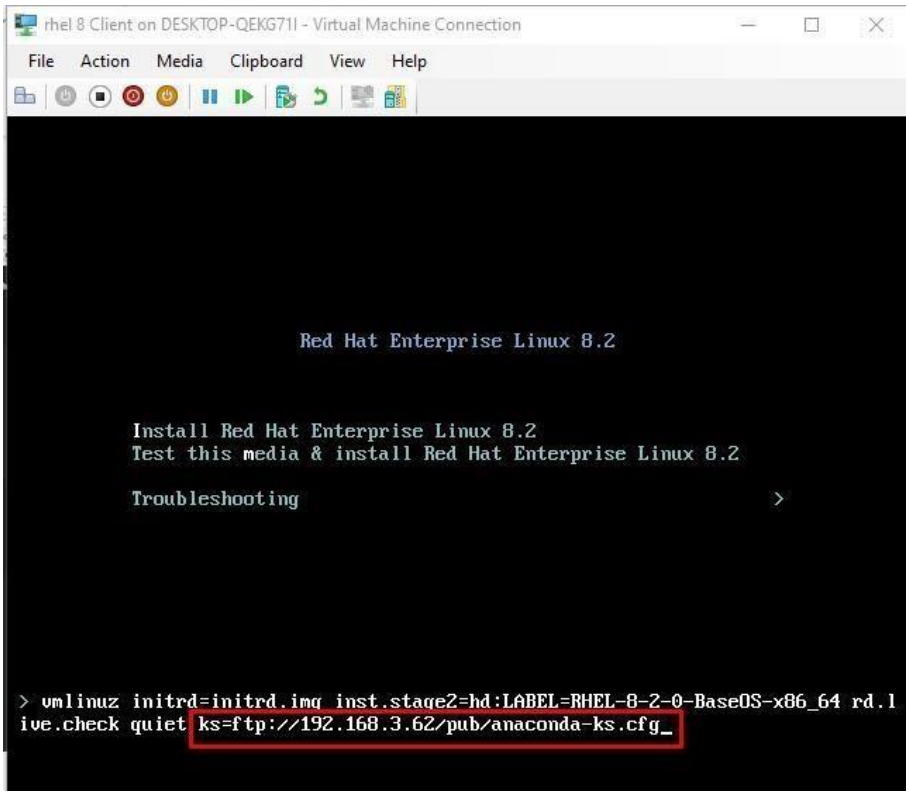
Configured the server with static IP **192.168.3.62**

Updated firewall rules to permanently allow ftp and dhcp services.

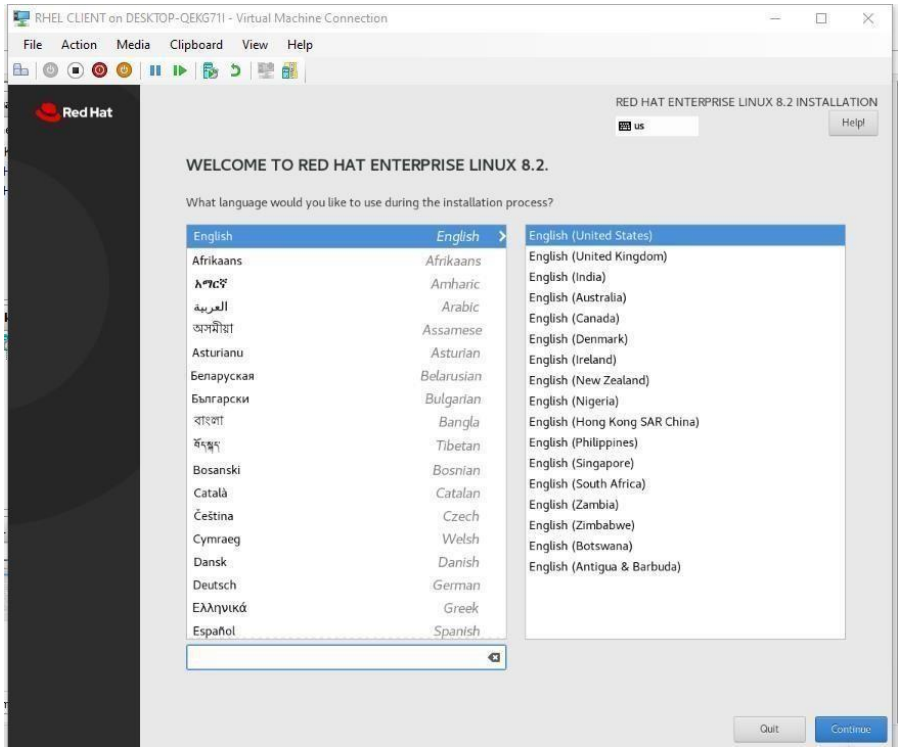
```
root@faculty:~
File  Edit  View  Search  Terminal  Help
Package vsftpd-3.0.3-36.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@faculty ~]# systemctl enable vsftpd
[root@faculty ~]# systemctl start vsftpd
[root@faculty ~]# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; vendor pres>
   Active: active (running) since Wed 2025-06-25 05:32:32 EDT; 19min ago
 Main PID: 1084 (vsftpd)
    Tasks: 1 (limit: 23084)
   Memory: 768.0K
   CGroup: /system.slice/vsftpd.service
           └─1084 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf

Jun 25 05:32:32 faculty systemd[1]: Starting Vsftpd ftp daemon...
Jun 25 05:32:32 faculty systemd[1]: Started Vsftpd ftp daemon.
[root@faculty ~]# firewall-cmd --add-service=ftp --permanent
Warning: ALREADY_ENABLED: ftp
success
[root@faculty ~]# firewall-cmd --add-service=dhcp --permanent
success
[root@faculty ~]#
```

Network Identification: The server's network interface eth0 was verified with the IP address 192.168.3.62 to serve as the installation source for client machines.



Client Validation: The client machine initiated the boot process, performed a 100% successful media integrity check, and successfully mounted the installation repository .



2. Virtualization and Management (KVM & Cockpit)

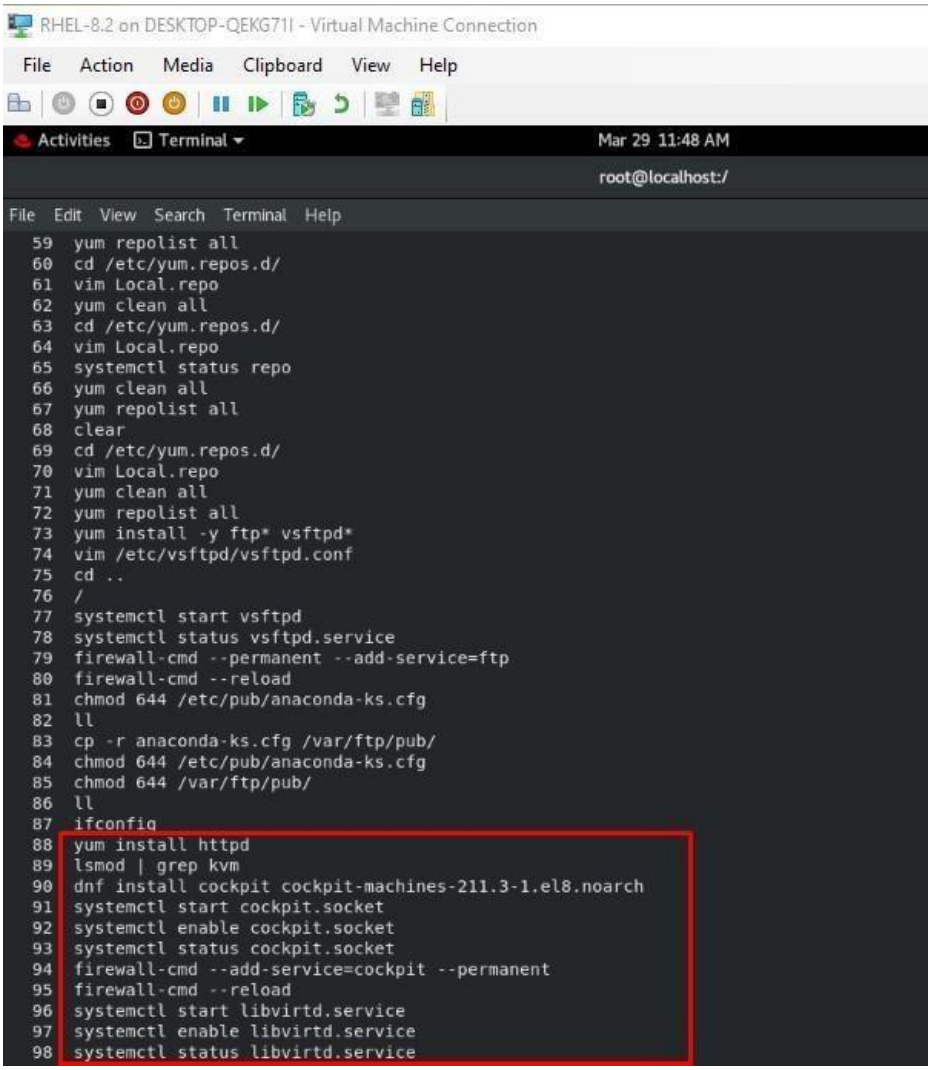
Objective

To implement a virtualization layer allowing the execution of multiple operating systems on a single physical RHEL 8 host.

Implementation Steps

- **Console Setup**: Installed cockpit and cockpit-machines to provide a web-based graphical interface for server management.

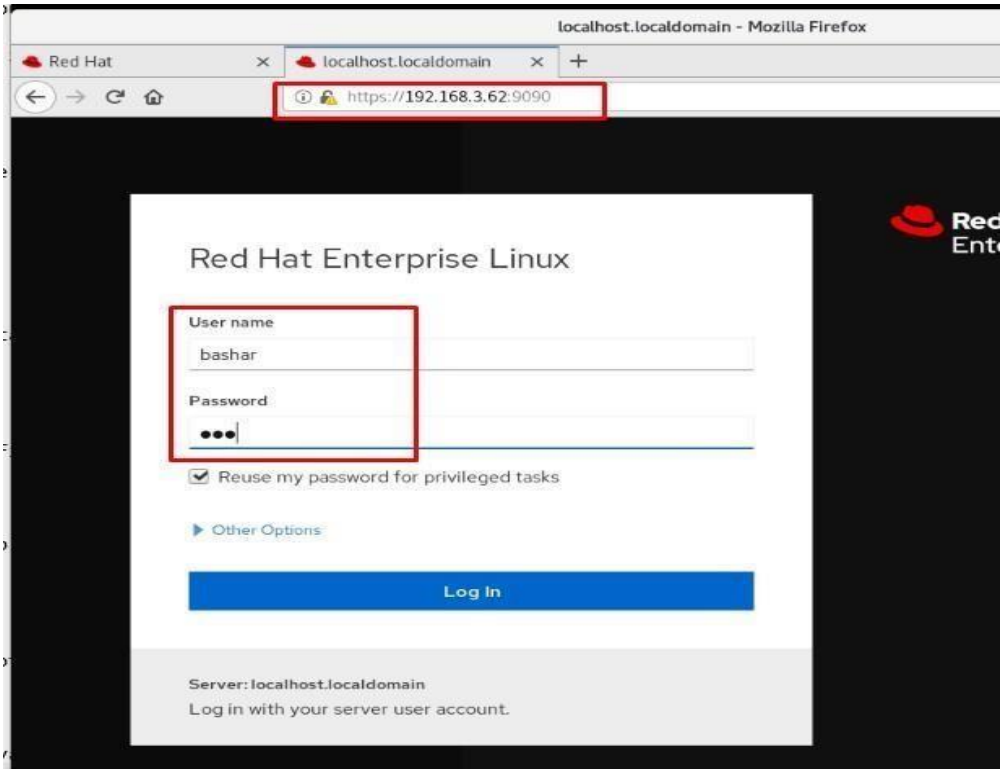
- **Virtualization Stack**: Enabled the **libvirtd.service** to manage the KVM (Kernel-based Virtual Machine) hypervisor.

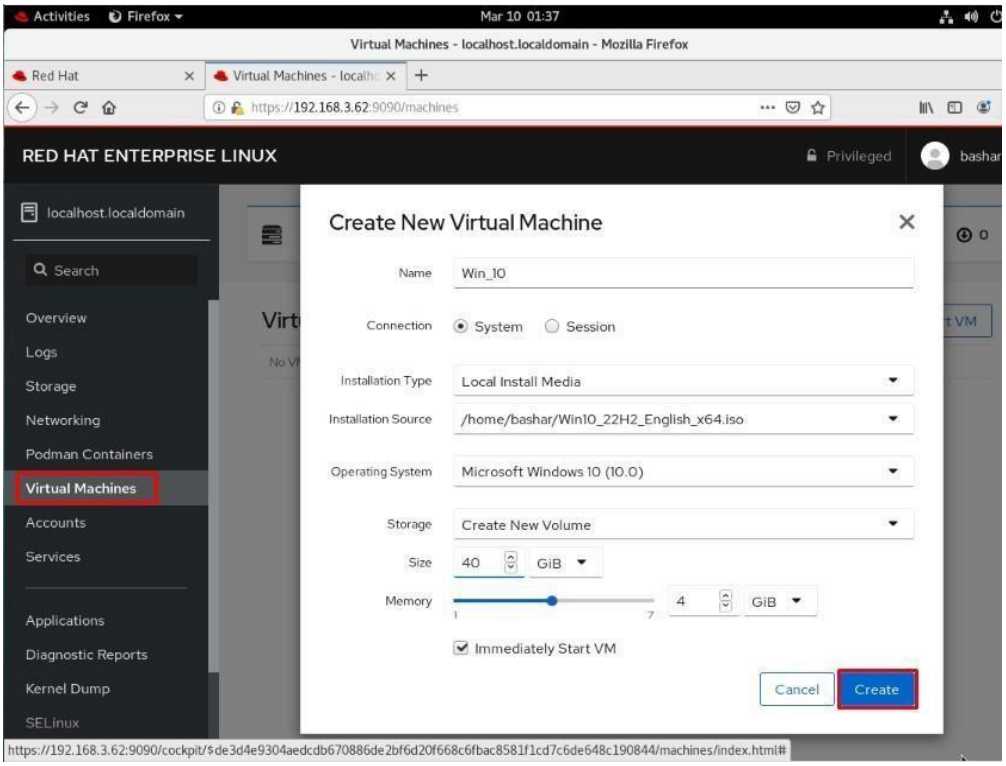


VM Provisioning:

- Accessed the management portal via https://192.168.3.62:9090.
- Created a virtual machine named **Win_10** with **4 GiB RAM** and **40 GiB Storage** using a

local ISO image.





3. Network File Sharing (Samba Server)

Objective

To facilitate seamless file sharing and interoperability between Linux and Windows systems within the network.

Implementation Steps :

- **Installation**: Deployed the samba, samba-common, and samba-client packages.

```
[root@localhost /]# yum install samba samba-common samba-client -y
Updating Subscription Management repositories.
Unable to read consumer identity
This system is not registered to Red Hat Subscription Management. You can use subscription-manager to r
egister.
Last metadata expiration check: 1:31:08 ago on Sun 29 Mar 2026 11:18:19 AM +06.
Package samba-common-4.11.2-13.el8.noarch is already installed.
Dependencies resolved.
==========================================================================================
=======================
 Package                          Architecture        Version                     Reposito
ry                Size
==========================================================================================
=======================
Installing:
 samba                            x86_64              4.11.2-13.el8               BaseOS
                  766 k
 samba-client                     x86_64              4.11.2-13.el8               BaseOS
                  658 k
Installing dependencies:
 samba-common-tools               x86_64              4.11.2-13.el8               BaseOS
                  472 k
 samba-libs                       x86_64              4.11.2-13.el8               BaseOS
                  170 k

Transaction Summary
==========================================================================================
=======================
Install  4 Packages

Total size: 2.0 M
Installed size: 5.8 M
Downloading Packages:
```

Storage Configuration:

- Created a dedicated sharing directory at /srv/samba/share/.

- Assigned ownership to nobody:nobody and set permissions to 775 for group collaboration.

Security (SELinux): Utilized semanage fcontext and restorecon to apply the samba_share t security context, ensuring the files are accessible through the Samba daemon.

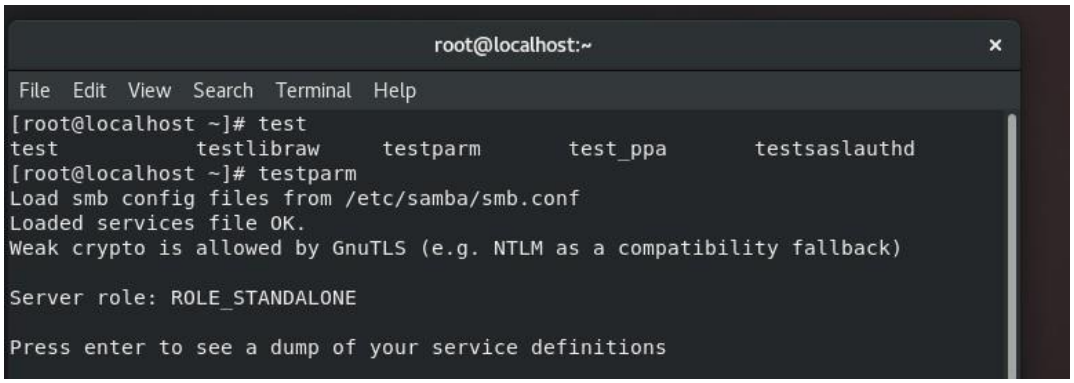
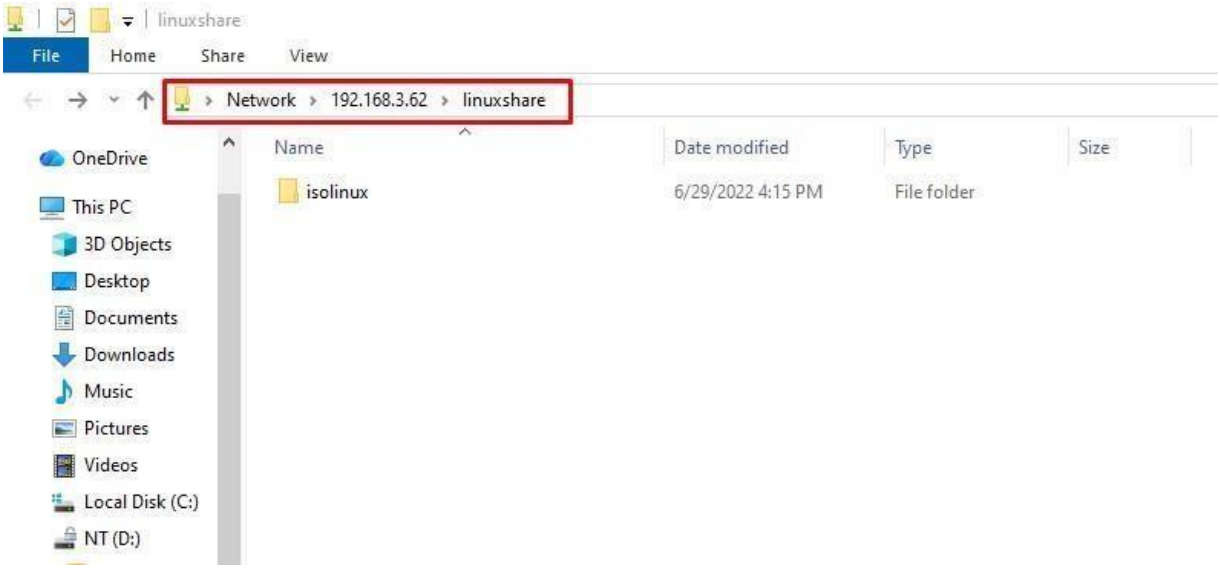
```
Activities  Terminal  Mar 29  1:06 PM
bashar@localhost:/
File  Edit  View  Search  Terminal  Help
  Verifying        : samba-libs-4.11.2-13.el8.x86_64
                   4/4
Installed products updated.

Installed:
  samba-4.11.2-13.el8.x86_64            samba-client-4.11.2-13.el8.x86_64       samba-common-tools-4.11
.2-13.el8.x86_64
  samba-libs-4.11.2-13.el8.x86_64

Complete!
[root@localhost /]# mkdir -p /srv/samba/share/
[root@localhost /]# chmod -R 775 /srv/samba/share/
[root@localhost /]# chmod -R 775 /srv/samba/share
[root@localhost /]# chown -R nobody:nobody /srv/samba/share/
[root@localhost /]# semanage fcontext -a -t samba_share_t "/srv/samba/share(/.*)?"
[root@localhost /]# restorecon -Rv /srv/samba/share/
Relabeled /srv/samba/share from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:samba_share_t:s
0
[root@localhost /]# restorecon -Rv /srv/samba/share
[root@localhost /]# nano /etc/samba/smb.conf
[root@localhost /]# smbpasswd -a root
New SMB password:
Retype new SMB password:
Added user root.
[root@localhost /]# systemctl start smb nmb
[root@localhost /]# systemctl enable smb nmb
Created symlink /etc/systemd/system/multi-user.target.wants/smb.service → /usr/lib/systemd/system/smb.s
ervice.
Created symlink /etc/systemd/system/multi-user.target.wants/nmb.service → /usr/lib/systemd/system/nmb.s
ervice.
[root@localhost /]# firewall-cmd --permanent --add-service=samba
success
[root@localhost /]# firewall-cmd --reload
success
[root@localhost /]#
```

```
66  yum install samba samba-common samba-client -y
67  mkdir -p /srv/samba/share/
68  chmod -R 775 /srv/samba/share/
69  chmod -R 775 /srv/samba/share
70  chown -R nobody:nobody /srv/samba/share/
71  semanage fcontext -a -t samba_share_t "/srv/samba/share(/.*)?"
72  restorecon -Rv /srv/samba/share/
73  restorecon -Rv /srv/samba/share
74  nano /etc/samba/smb.conf
75  smbpasswd -a root
76  systemctl start smb nmb
77  systemctl enable smb nmb
78  firewall-cmd --permanent --add-service=samba
79  firewall-cmd --reload
80  histor
81  history
[root@localhost /]#
```

testparm is a command-line utility used to verify the correctness of Samba (smb.conf) configuration files.



Validation: Verified connectivity by mapping the network drive \\192.168.3.62\linuxshare from a Windows client.